



Purpose

To familiarize with common laboratory techniques and apply critical thinking to measurement data and chemical reaction schemes. We will address these questions:

- Which procedure for measuring density yields the most accurate and precise data?
- What is the metal object made of?
- What is the identity of the unknown ionic solution?

Laboratory Activities

- A. Physical Analysis I - Precision and Accuracy of Technique
- B. Chemical Analysis - Identifying Ionic Solutions by Reaction Observations
- C. Physical Analysis II - Identifying Metal Objects by Density
- D. Group Argumentation and Discussion

Laboratory Skills

Weighing solids
Measuring volume
Recording qualitative and quantitative observations
Vacuum filtration of solids

Introduction

Watch the [introductory video](#) for this experiment.

In this experiment, we will investigate the composition of two different categories of matter: metallic solids and ionic aqueous solutions. We will study the impact of the accuracy and precision of a laboratory technique on the quality of the data we obtain and the confidence we have in interpreting our results. We will implement critical thinking and draw logical conclusions based on both evidence and theory. Throughout the experiment, careful technique is required to obtain accurate and precise results. Take some time in the laboratory to become comfortable with the tools at your disposal. Developing good habits now will make you more accurate, precise, and efficient in future experiments.

Measurements, Units, and Significant Digits

We will make use of the SI system of units throughout CHEM 1310. The seven fundamental quantities in this system are listed in Table LF.1 along with the *base units*.

Table LF.1: SI System of Units

Physical Quantity	Unit	Abbreviation
mass	kilogram	kg
length	meter	m
time	second	s
electric current	ampere	A
temperature	Kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd

The base units can be combined to form *derived units* with different names, such as the Newton (1 kilogram-meter per second squared) and Coulomb (1 ampere-second). Prefixes are added before the name of a base or derived unit to represent scaling by a power of ten. Familiarize yourself with the abbreviations and scaling factors associated with each prefix shown in Table LF.2. When making measurements in the laboratory, carefully note the units of each instrument or piece of glassware that you use.

Occasionally, you will need to convert a measurement from one unit to another. This can happen, for example, when you have a physical constant in one unit but measurements were made in an incompatible unit. Unit conversions may also be necessary when comparing values (for instance, in the assessment of accuracy). *Conversion factors* are ratios that enable the transformation of a measurement from one set of units to another. The magnitude of a conversion factor depends on the relative sizes of the two units involved.

In making measurements and performing calculations based on these measurements, you should also carefully consider *significant digits*. Measuring instruments can only provide a finite level of precision; at some point, the size of a measurement becomes imprecise or “fuzzy.” When using a digital measuring instrument such as the LabQuest, record all of the digits that the instrument gives you, estimating the last digit if some “wobble” or drift is observed. Digits after the smallest place reported on the device are uncertain and cannot be estimated. All of the digits in this measurement are considered significant, even trailing zeroes.

Analog instruments and glassware are similarly limited in their precision. When we refer to the precision of a piece of glassware or instrument, we mean the smallest amount that can be measured with a reasonable amount of certainty. For laboratory glassware, it is possible to systematically determine precision using the markings on the glassware. The general rule here is that one and only one digit can be estimated; other digits come from numbers printed on the glassware. Again, all of the digits in these measurements are considered significant,

Table LF.2: Scientific Prefixes

Prefix	Abbreviation	Scaling Factor
femto	f	10^{-15}
pico	p	10^{-12}
nano	n	10^{-9}
micro	μ	10^{-6}
milli	m	10^{-3}
centi	c	10^{-2}
deci	d	10^{-1}
kilo	k	10^3
mega	M	10^6
giga	G	10^9
tera	T	10^{12}

but it will be up to you to record the measurement with the correct number of significant digits. If you think a measurement is sitting right on a marking, add a trailing zero!

When performing calculations using measurements, it is important to avoid adding false precision to calculated values. Such false precision could lull a reader into a false sense of security about the quality of the measurements. Be sure to apply the rules for mathematical operations with significant digits in all calculations in CHEM 1310 laboratory. With practice, the rules should become highly intuitive and you'll be able to breeze through complex calculations while keeping a mental inventory of significant digits.

Accuracy and Precision

Empirical measurements represent the basis of everything that chemists know. More generally, any scientific understanding of the world is grounded in measurements. Measurement theory clarifies the *limitations* of scientific measurement and gives us tools to evaluate the quality of data. You'll find these tools useful not just in your time as a student in CHEM 1310 laboratory, but also throughout your career as a scientist or engineer.

Data quality has two independent dimensions: precision and accuracy. *Accuracy* is defined as the closeness of a measurement or set of measurements to a true, theoretical, or previously measured value. Although naively we might quantify accuracy as the difference between the mean measured value and the true value, this difference doesn't mean much without information about the magnitude of the true value. (Is 3 grams a lot? If the true value is 4 grams, yes; if the true value is 500 grams, no.) Thus, we divide this difference by the true value and multiply by 100% to obtain the key measure of accuracy, *percent error*.

$$\text{PE} = \left| \frac{x_{\text{mean}} - x_{\text{true}}}{x_{\text{true}}} \right| \times 100\% \quad (\text{Equation LF.1})$$

Precision is defined as the spread within a set of equivalent measurements. Measurements that are very similar to one another are said to be precise; measurements that are spread out over a large range are said to be imprecise. The key measure of precision is *standard deviation* (s). Defining $\bar{x}$ as the mean of the data and N as the number of measurements,

$$s = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2} \quad (\text{Equation LF.2})$$

Keep in mind that precision applies only to a set of equivalent measurements, as when the same experiment is run multiple times. Accuracy, on the other hand, can be evaluated for any number of measurements (to evaluate the accuracy of a set of equivalent measurements, the mean of the set is used).

Solubility and Precipitation Reactions

When placed in water, many ionic solids or salts dissolve and undergo dissociation. On the microscopic level, *dissolution* involves the surrounding of the ionic compound by water molecules, which interact with the ions through electrostatic forces. During *dissociation*, the cations and anions in the ionic solid separate as they are individually surrounded by water molecules. Under the right circumstances, dissolution can happen in reverse: cations and anions can form ionic bonds and leave solution, forming an ionic solid (*precipitation*). Precipitation occurs in any solution containing a greater concentration of ions than the maximum solubility of the salt. We can separate salts roughly into three groups depending on the magnitude of their solubility: *soluble* salts have very large solubility, *sparingly soluble* salts have moderate solubility, and *insoluble* salts have very small solubility. When the cationic and anionic components of an insoluble salt find themselves in the same solution, they will precipitate spontaneously.

For example, when an aqueous solution of lead(II) nitrate, $\text{Pb}(\text{NO}_3)_2$, is combined with an aqueous solution of potassium iodide, KI, a precipitate forms that consists of lead(II) iodide, PbI_2 . The $\text{Pb}(\text{NO}_3)_2$ and KI solutions are prepared by dissolving the corresponding ionic solids in water, as both salts are soluble. However, when the solutions are combined, Pb^{2+} and I^- ions find themselves in the same solution, and PbI_2 is insoluble in water. As a result, PbI_2 forms spontaneously in the mixture:



The *solubility rules* are qualitative guidelines that enable us to predict the solubility of a salt from its component cation(s) and anion(s). The rules list groups of salts that are soluble or insoluble, with exceptions associated with specific cations or anions.

1. Salts of group 1 cations (Li^+ , Na^+ , and K^+) are soluble.
2. All common acetates (CH_3CO_2^-) and nitrates (NO_3^-) are soluble.
3. All chlorides, bromides, and iodides are soluble with the exception of salts of silver, lead(II), and mercury(I).
4. Sulfates (SO_4^{2-}) are soluble with the exception of salts of barium, strontium, lead(II), calcium, silver, and mercury(I).
5. Carbonates (CO_3^{2-}), hydroxides (OH^-), oxides (O^{2-}), and phosphates (PO_4^{3-}) are insoluble, except for salts of group 1 cations.

In this experiment, we will determine the identity of a salt solution using small-scale precipitation tests. The unknown solution will be mixed systematically with known solutions and observations of the precipitate formed (if any) will be recorded. Combining the observations with the solubility rules allows us to deduce the cation and anion in the unknown solution. The cation may be barium, silver, sodium, or copper(II) and the anion may be chloride, nitrate, or sulfate.

Procedures

Reagents Available

- Sodium sulfate (Na_2SO_4) solution, 0.1 M, [Safety for Sodium Sulfate Solution](#)
- Lead(II) nitrate ($\text{Pb}(\text{NO}_3)_2$) solution, 0.1 M, [Safety for Lead\(II\) Nitrate Solution](#)
- Sodium hydroxide (NaOH) solution, 0.1 M, [Safety for Sodium Hydroxide Solution](#)
- Sodium chloride (NaCl) solution, 0.1 M, [Safety for Sodium Chloride Solution](#)
- Barium nitrate ($\text{Ba}(\text{NO}_3)_2$) solution, 0.1 M, [Safety for Barium Nitrate Solution](#)
- Unknown solutions **A – D** (salts of Cu^{2+} , Ba^{2+} , Ag^+ , or Na^+ along with Cl^- , NO_3^- , or SO_4^{2-})

Safety Note

Remember that glass is brittle and is liable to break when dropped. Keep glassware away from the edges of your workspace and lift glassware whenever possible to make measurements. Don't lower your head to make a measurement—this leaves you more vulnerable to spills.

The ionic salt solutions, particularly those of barium and lead(II), are associated with environmental hazards. To minimize the risks of these hazards, manage the waste from the precipitation tests responsibly. Discard the liquid waste from this experiment in the designated waste bottle. **First**, check the liquid level of the bottle to ensure that it isn't more than 75% full (if it is, let your teaching assistant know that a waste pickup is needed). Then, lift the red lid of the safety funnel and pour the waste into the bottle. Finally, replace the lid, which minimizes the escape of vapors from the bottle.

Part A. Physical Analysis I

Goal: Determine the density of each of 6 metal cubes.

Research Question: Which procedure for measuring density yields the most accurate and precise data?

- Each group will be assigned one of the following combinations of techniques to define your procedure for measuring and calculating the cubes' densities.

A Ruler + Top Loader Balance	B Caliper + Top Loader Balance	C Water Displacement + Top Loader Balance
D Ruler + Analytical Balance	E Caliper + Analytical Balance	F Water Displacement + Analytical Balance

- Refer to the instructions relevant to your group's combination of techniques and develop a plan for efficiently collecting all the necessary data (mass in grams, volume in cm^3 for all 6 blocks). To collect 3 trials of each, allow 3 team members to make the measurements for all blocks according to your procedure (one of A-F).
- Technique Instructions:**
 - **Ruler:** Measure each of the 3 dimensions (length, width, height) of the block to the smallest decimal place allowed based on the markings of the ruler (estimate one decimal place beyond the markings). To obtain densities in g/cm^3 , measurements should be made in units of centimeters.

- **Caliper:** Slide the caliper open and place the block between the two points and slide closed until the points are making full contact with the sides of the block. Record the length, width, and height of the block in centimeters to the maximum number of decimal places provided by the device.
 - **Water Displacement:** Fill a graduated cylinder wide enough to hold the block inside with water. Record the starting volume of water inside to one estimated decimal place beyond the markings provided on the graduated cylinder. Submerge the block in the water and record the final volume of water. This will result in a volume in terms of milliliters which can easily be converted into cm^3 .
 - **Top Loader Balance:** Check that the top of the balance is clean and that the balance is turned on. Zero (tare) the balance prior to placing anything on top. Record the mass of the block including all decimal places provided.
 - **Analytical Balance:** Check that the top of the balance is clean and that the balance is turned on. Zero (tare) the balance prior to placing anything on top. Open the door and place the block on the balance plate. Record the mass of the block including all decimal places provided.
4. Calculate densities to an appropriate number of significant figures by following significant figures rules:
- When measured values are multiplied or divided, the resulting value should be rounded to match the least number of significant figures from the measured values.
 - Example: 0.2340 (4 sf) divided by 0.00354 (3 sf) = $66.10169 \longrightarrow$ rounded: **66.1** (3 sf)
 - When measured values are added or subtracted, the resulting value should be rounded to match the least number of decimal places in the measured values.
 - Example: 0.1784 (4 dp) minus 0.083 (3 dp) = 0.0954 (4 dp) $\longrightarrow$ rounded: **0.095** (3 dp)

Table LF.3: Data for Known Metal Blocks

Group Iron Block	Proce- dure A	Proce- dure B	Proce- dure C	Proce- dure D	Proce- dure E	Proce- dure F
Mass 1 (g)						
Mass 2 (g)						
Mass 3 (g)						
Volume 1 (cm^3)						
Volume 2 (cm^3)						
Volume 3 (cm^3)						

5. All groups will combine their density data for the 6 blocks in a class data table indicating which combination of techniques was used. This class data will be needed for the post-lab assignment.

Table LF.4: Average Density Class Data

Group Metal	Proc. A	Proc. B	Proc. C	Proc. D	Proc. E	Proc. F
Aluminum						
Brass						
Copper						
Iron						
Lead						
Zinc						

6. Using the labeled identities of the materials comprising each block, determine which procedure yielded the most accurate and precise data set based on all available evidence.

Refer to Table LF.5 for some densities of common metals..

Table LF.5: Densities of Common Metals

Metal	Density (g/cm ³)	Metal	Density (g/cm ³)
Iron	7.87	Gold	19.3
Silver	10.5	Nickel	8.90
Lead	11.34	Titanium	4.51
Tin	7.26	Chromium	7.15
Aluminum	2.70	Bronze	8.7
Steel	7.86	Mercury	13.53
Copper	8.96	Zinc	7.14
Brass	8.5	Magnesium	1.74

Part B. Chemical Analysis

Goal: Identify an unknown ionic solution based on solubility rules.

Research Question: What is the identity of the unknown ionic solution?

1. Make a prediction about the results of each of the reactions in the table in your report sheet by writing the products and indicating if a precipitate will form.

Table LF.6: Part B Hypotheses

Reactants	Products	Precipitate? (yes/no & formula)
$\text{Ba}(\text{NO}_3)_2 (\text{aq}) + \text{NaCl} (\text{aq})$		
$\text{Pb}(\text{NO}_3)_2 (\text{aq}) + \text{NaCl} (\text{aq})$		
$\text{Ba}(\text{NO}_3)_2 (\text{aq}) + \text{Na}_2\text{SO}_4 (\text{aq})$		
$\text{Pb}(\text{NO}_3)_2 (\text{aq}) + \text{Na}_2\text{SO}_4 (\text{aq})$		

2. Test the hypotheses made above based on theory with experimental evidence. Obtain a ceramic spot plate or well plate. Place a few drops of each reactant into distinct wells to observe their reaction. Record your observations.

3. Generate a new set of hypotheses for determining the identity of your unknown using the table provided in the report sheet. Indicate whether a precipitate will be present for each combination.

Table LF.7: Part B New Hypotheses

Solutions	Na_2SO_4 (aq)	$\text{Pb}(\text{NO}_3)_2$ (aq)	NaOH (aq)	NaCl (aq)	$\text{Ba}(\text{NO}_3)_2$ (aq)
Cu^{2+} (aq)					
Ba^{2+} (aq)					
Ag^+ (aq)					
Na^+ (aq)					
Cl^- (aq)					
NO_3^- (aq)					
SO_4^{2-} (aq)					

4. Select an unknown solution (**A – D**) and record its letter. Place a few drops of the unknown into each of 5 separate wells. Add each of the 5 known solutions to separate wells to react with the unknown solution. Record observations
5. Carefully pour the reaction mixtures from your spot plate into a waste beaker, using your wash bottle with water to rinse off any precipitates. Ensure that there is enough water in the beaker to transfer the solid. We will vacuum filter this mixture to separate the solid and liquid phases.
6. Set up a vacuum filtration apparatus by connecting a Büchner funnel to a filter flask. Use a ring stand and clamp to hold the flask in place, then use vacuum tubing to connect the nozzle of the flask to a vacuum tap at your bench.
7. Weigh a piece of filter paper, record its mass, and add the filter paper to the Büchner funnel.
8. Wet the filter paper with deionized water and turn on the vacuum to ensure a good seal.
9. Gently pour the mixture of solid and liquid over the filter paper. Once all of the liquid has been pulled through, allow the vacuum to pull air through the solid for 5 additional minutes to help it dry.

10. Turn off the vacuum, and carefully remove the filter paper with the solid.
11. Weigh the filter paper with the solid on it and record the paper + solid mass. Use subtraction to determine the mass of solid obtained.
12. Rinse the liquid in the filter flask into the hazardous liquid waste bottle in the hood. Use a scoopula to scrape the solid off of the filter paper and into the solid waste container. Discard the filter paper in the trash.

Part C. Physical Analysis II

Goal: Identify the metal in an unidentified object.

Research Question: What is the metal object made of?

1. Select an unidentified metal object to analyze. Write a short description of the object.
2. Based on findings for the most accurate and precise procedure for measuring density from Part A, as well as the shape/size of the object you selected, determine the best procedure for analyzing the density of your object. Write down the plan before moving on.
3. Carry out your experiment and determine the density of the object to the correct number of significant figures based on 3 trials.
4. Return to the research question for Part C. Construct an argument for the identity of the metal, list appropriate evidence used in drawing that conclusion, and provide reasoning by explaining how the selected evidence supports the conclusion.

D. Group Argumentation and Discussion

With the other students at your bench, compile class and/or group data and analyze any trends. Return to the research question and arrive at a conclusion addressing the question together. Generate an argument from evidence that can be used in your post-lab report. Record the trends and argument in your lab notebooks before leaving.

Post-lab Calculations and Data Workup

Post-lab Part A:

Enter class data into a table. Determine which combo of techniques results in the most precise measurement. Calculate mean and standard deviation from class data. Interpret these values to identify the most accurate and precise set of data.

Post-lab Part C:

Calculate the average density of unknown object and determine the identity of the metal. Calculate the percent error of density.

Write-Up Instructions

The post-lab assignment for this experiment is an abbreviated lab report for argument-driven inquiry (ADI) experiments following the structure in the provided template.

All tables, figures, and writing should meet the minimum expectations described in the science writing resources.

ADI Report (Introduction, Methods, Results, and Conclusion)

Follow the prompts in the general report template to structure your paper.

- A brief background on what you were trying to do that ties to any relevant theoretical background knowledge is intended to lead into your communication on your findings.
- The methods section is a very brief explanation of how you obtained your data and analyzed it. - To complete the Results section, copy and paste important figures and tables from above into the document. Brief text describing the results is important in this section. Include relevant calculations (a sample of each type of calculation). Construct equations using the Equation Editor built into Microsoft Word or an online equation editor. If you use an online editor, include equations as images, not text.
- To complete the Discussion for this experiment, address the following items in paragraph form. Do not include the prompts in your submission; weave your answers together into a coherent narrative.
 1. Make a claim about which set of techniques yielded the most precise and accurate data and which resulted in the least accurate and precise data. Provide evidence for these claims including the statistical analyses of the density values and explain this evidence as it relates to either accuracy or precision.
 2. Write an argument that identifies the metal in the selected unknown object considering the error in the measurement and any uncertainty in the evidence collected. Assess the effectiveness of the procedure utilized for this density measurement based on the results.
 3. Provide a rationalization of your conclusion concerning the identity of the unknown salt solution. Identify chemical formulas or names for all precipitates formed and demonstrate logically how your results imply the cation and anion of your proposed salt.